DR BEN MASLEN

Ph: +61438 790 990 Email: b.maslen@unsw.edu.au

Research Website LinkedIn Google Scholar Github ORCID

PROFESSIONAL SUMMARY

Interdisciplinary statistician and data scientist with a PhD in Statistics from the University of New South Wales and CSIRO Data61. I develop statistical and artificial intelligence methods that help ecologists monitor and protect the natural environment, with expertise in statistical methodology, ecological statistics, and applied data science. I have a passion for teaching and communicating statistics to diverse audiences.

EXPERIENCE

LECTURER, 2026, JUL - CURRENT
SCHOOL OF MATHEMATICS AND STATISTICS, UNIVERSITY OF NEW SOUTH WALES

Lectured and co-convened the statistics component of Civil and Environmental Engineering Computations (CVEN2002) and currently lecturing Statistics for Life and Social Sciences (MATH1041), responsible for delivering and coordinating statistics teaching for over 800 first and second-year students. My research is at the intersection of ecology, statistics, and data science, as well as interdisciplinary applied research through ongoing collaborations with researchers at UNSW, Wentworth Group of Concerned Scientists, Prince of Wales Hospital and Taronga Zoo.

STATISTICAL CONSULTANT, 2021, JUL - 2026, JUN &
MARK WAINWRIGHT ANALYTICAL CENTRE, UNIVERSITY OF NEW SOUTH WALES 2018, SEP - 2019, DEC

Collaborated with over 220 academics, researchers, and higher degree research students at the University of New South Wales and partner institutions from data science, plant, marine, and terrestrial ecology, community health, medical research, pharmacology, engineering and genome research fields to provide appropriate design and analysis advice relative to their research questions and data properties. I also regularly prepared content and facilitated seminars and courses on coding and statistical analysis methods for researchers and higher degree research students. Through this role, I was contracted by the School of Mathematics and Statistics to supervise multiple master's student groups taking the Data Science and Decisions Project course (DATA5925 & MATH5925), guiding students through an independent research project tackling real world data science problems (2026).

DATA SCIENTIST, 2026, JAN - 2026, FEB,
WENTWORTH GROUP OF CONCERNED SCIENTISTS 2020, JAN - 2020, JUN

Applied statistical analysis to environmental policy questions for the Wentworth Group of Concerned Scientists, investigating changes in the outward flow and reliability of water shares under a drier climate in the Murray-Darling Basin and land clearing in NSW following 2016 environmental reforms.

STATISTICAL INSIGHTS ANALYST, 2020, SEP - 2021, DEC
HEALTHDIRECT

Developed reporting and advanced modelling insights to enhance the management of health and user experience data across a range of Healthdirect services.

DATA ANALYST, 2018, MAY - 2019, MAY
NATURAL RESOURCE ACCESS REGULATOR, NSW GOVERNMENT

Developed and implemented a reporting and data management system for the NSW Natural Resource Access Regulator. This enabled the tracking of workflow for the strategic management and triaging of tasks.

MATHEMATICS AND STATISTICS DEMONSTRATOR, 2018, FEB - 2018, MAY
SCHOOL OF MATHEMATICS AND STATISTICS, UNIVERSITY OF NEW SOUTH WALES

Academic demonstrator for mathematics and statistics courses including; Higher Probability and Stochastic Processes, Numerical Methods and Statistics, and Mathematicss for Life and Social Sciences.

EDUCATION

PHD IN STATISTICS, 2021-2025
UNIVERSITY OF NEW SOUTH WALES & CSIRO DATA61

During my PhD, I developed and applied advanced AI techniques and statistical modelling to improve the analysis of ecological video data. I enhanced AI object detection by incorporating animal movement patterns, evaluated the reliability of commonly used ecological metrics, and implemented simulation-based approaches to assess bias and statistical power. I also adapted distance sampling methods to estimate fish density, enabling a novel approach to gain accurate ecological insights from imperfect observational data.

HONOURS OF STATISTICS (1ST CLASS), 2017
UNIVERSITY OF NEW SOUTH WALES

BACHELOR OF ADVANCED SCIENCE (HIGH DISTINCTION), 2014-2016
UNIVERSITY OF NEW SOUTH WALES

TEACHING EXPERTISE

LECTURING:

- Statistics for Life and Social Sciences, (School of Mathematics and Statistics, UNSW) 2026
- Civil and Environmental Engineering Computations, (School of Mathematics and Statistics, UNSW) 2026
- Introduction to Mixed Models with R (Stats Central, UNSW) 2026
- Statistics for Medical Science (Stats Central, UNSW) 2025-2026
- Introduction to Regression with R (Stats Central, UNSW) 2019, 2021-2025
- Introduction to R (Stats Central, UNSW) 2019, 2022
- Introduction to Statistics with R (Stats Central, UNSW) 2022
- Ecological statistics (Stats Central, UNSW) 2018-2019, 2021-2022

ACADEMIC DEMONSTRATOR:

- Higher Probability and Stochastic Processes (School of Mathematics and Statistics, UNSW) 2018
- Numerical Methods and Statistics (School of Mathematics and Statistics, UNSW) 2018
- Mathematics for Life and Social Sciences (School of Mathematics and Statistics, UNSW) 2018

SUPERVISION:

School of Mathematics and Statistics, UNSW (2026) – Supervisor for two groups of Masters students taking the Data Science and Decisions Project course (DATA5925 & MATH5925), guiding students to independently research and analyse a VISA tribunal outcomes dataset, tackling real world data science problems.

Stats Central, UNSW (2018-2019, 2021-2026) – I have met and helped with the supervision of over 180 honours and postgraduate research students, guiding them through the experimental design and analysis phase of their research projects in collaboration with their academic supervisors.

TECHNICAL AND CODING SKILLS

- **R** (Statistical Computing Language)
- **Python (**General Purpose Programming Language)
- **SQL** (Data Management Language)
- **SPSS** (Statistical Computing Software)
- **Matlab** (Technical Computing Language)
- **Bash/Linux** (High Performance Computing)

PUBLICATION RECORD

Peer reviewed research articles: 17 (12 **Q1**, 5 **Q2**)

Total citations: 341 **H index**: 8 **i10 index**: 7

Publications of interest:

The Motion Picture: Leveraging Movement to Enhance AI Object Detection in Ecology 2025

This study evaluated frame differencing, background subtraction, optical flow and tracking methods integrated into a YOLOv8 AI object-detection pipeline in Python. Across more than 35,000 annotated images from terrestrial, marine and freshwater habitats, we found movement information was especially valuable for smaller studies and rarer species. The implementation is available through a public GitHub repository, with a public benchmark dataset released through Dryad.

How many sites? Methods to assist design decisions when collecting multivariate data in ecology 2023

This paper developed a power-analysis framework for multivariate ecological data combining Gaussian copula and factor-analytic models, expert-informed effect sizes and an efficient critical-value approach to estimate power. The accompanying open-source ecopower R package makes the method accessible to ecologists and has been downloaded more than 14,000 times from CRAN.

ACHIEVEMENTS AND PRESENTATIONS

Invited Speaker:

- Australian Statistical Conference, Perth, Australia 2025
  - Mentor for statistical consulting session

Contributed Speaker:

- Australian Fish Biology Conference, Darwin, Australia 2025
- International Statistical Ecology Conference, Swansea, Wales 2024
- UNSW Mathematics and Statistics Postgraduate Conference, Sydney, Australia 2023, 2024
  - Peoples choice award for best presentation (2024)
- UNSW 1 minute thesis competition 2022
  - 1st place in the School of Mathematics and Statistics
- Statistical Ecology and Environmental Monitoring Conference, Queenstown, New Zealand 2017

Scholarships:

- Australian Government Research Training program scholarship 2021-2024
- CSIRO, Data61 PHD top up scholarship 2021-2024
- UNSW School of Mathematics and Statistics PHD top up scholarship 2021-2024

Awards:

- Tjanpi Award for best Student paper in Environmental Statistics (runner up) 2023

ACADEMIC REFEREES

- Professor David Warton (PhD supervisor & colleague, School of Mathematics and Statistics, UNSW) - david.warton@unsw.edu.au, +61 2 9385 7031
- Dr Nancy Briggs (manager at Stats Central) - nancy.briggs@unsw.edu.au, +61 2 9385 8220
- Professor Jake Olivier (Head of Statistics, School of Mathematics and Statistics, UNSW) - j.olivier@unsw.edu.au +61 2 9385 6656
- Dr Peter Humburg (colleague at Stats Central) - peter.humburg@unsw.edu.au, +61 2 9385 8220

PUBLICATIONS

Anderson, S., Norden, S., Maslen, B., & Baitch, L. (2026). Frailty and time-to-theatre for elderly hip fracture patients at a regional hospital in australia: A retrospective cohort study. *Geriatric Orthopaedic Surgery & Rehabilitation*, *17*, 21514593261487222. https://doi.org/10.1177/21514593261487222

Ignacio, B. L., Moles, A. T., Offord, C. A., Chiarenza, G. M., Maslen, B., Hemmings, F. A., Leggett, K., & Xirocostas, Z. A. (2026). Soil seed bank holds some potential for natural regeneration in a semi-arid rangeland. *Austral Ecology*, *51*(8), e70278. https://doi.org/10.1111/aec.70278

Langley, M., Vergés, A., Maslen, B., Lynch, T., & Poore, A. (2026). Remote underwater video and analysis with AI show differences between fish communities at chain moorings and at environmentally friendly moorings. *Marine & Freshwater Research*, *77*(3). https://doi.org/10.1071/MF25208

Lau, S. M., Popovic, G., Maslen, B., & Depczynski, B. (2026). Early, unplanned readmission in diabetes inpatients: Comparing the utility and performance of machine learning and traditional prediction models to guide timely diabetes management team review. *Diabetic Medicine*, *43*(2), e70186. https://doi.org/10.1111/dme.70186

Ford, A. M., Grootemaat, S., Towers, I., Maslen, B., Cornwell, W. K., Ooi, M. K., & Falster, D. S. (2025). Assessing the effect of tissue and fire-response traits on plant growth rates post-disturbance in eastern australia. *Functional Ecology*, *39*(1), 320–332. https://doi.org/10.1111/1365-2435.14698

Maslen, B. (2025). *Swimming in data: Using AI and distance sampling to improve the analysis of remote underwater videos in estimating fish abundance* [PhD thesis, University of New South Wales (Australia)]. https://doi.org/10.26190/unsworks/31635

Maslen, B., Popovic, G., Wang, D., Jansen, A., & Warton, D. (2025). The motion picture: Leveraging movement to enhance AI object detection in ecology. *Ecology and Evolution*, *15*(8), e71996. https://doi.org/10.1002/ece3.71996

Coleman, A., Maslen, B. J., & Foster, R. (2024). Inequities in PrEP use according to medicare status in a publicly funded sexual health clinic; a retrospective analysis. *Sexual Health*, *21*(2), SH23141. https://doi.org/10.1071/SH23141

Maslen, B., Popovic, G., Lim, M., Marzinelli, E., & Warton, D. (2023). How many sites? Methods to assist design decisions when collecting multivariate data in ecology. *Methods in Ecology and Evolution*, *14*(6), 1564–1573.https://doi.org/10.1111/2041-210X.14094

Brandenburger, C. R., Maslen, B., Sherwin, W. B., & Moles, A. T. (2022). Weedy and seedy: The rapid evolution of life-history characteristics in an introduced daisy. *AoB Plants*, *14*(5), plac038. https://doi.org/10.1093/aobpla/plac038

Kirubakaran, R., Hennig, S., Maslen, B., Day, R. O., Carland, J. E., & Stocker, S. L. (2022). Evaluation of published population pharmacokinetic models to inform tacrolimus dosing in adult heart transplant recipients. *British Journal of Clinical Pharmacology*, *88*(4), 1751–1772. https://doi.org/10.1111/bcp.15091

Muskovic, W., Slavich, E., Maslen, B., Kaczorowski, D. C., Cursons, J., Crampin, E., & Kavallaris, M. (2022). High temporal resolution RNA-seq time course data reveals widespread synchronous activation between mammalian lncRNAs and neighboring protein-coding genes. *Genome Research*, *32*(8), 1463–1473. https://doi.org/10.1101/gr.276818.122

Khan, S., Schilling, H. T., Khan, M. A., Patel, D. K., Maslen, B., & Miyan, K. (2021). Stock delineation of striped snakehead, channa striata using multivariate generalised linear models with otolith shape and chemistry data. *Scientific Reports*, *11*(1), 8158. https://doi.org/10.1038/s41598-021-87143-9

Leroy, E. C., Royer, J.-Y., Alling, A., Maslen, B., & Rogers, T. L. (2021). Multiple pygmy blue whale acoustic populations in the indian ocean: Whale song identifies a possible new population. *Scientific Reports*, *11*(1), 8762. https://doi.org/10.1038/s41598-021-88062-5

Mesaglio, T. P., Schilling, H. T., Adler, L., Ahyong, S. T., Maslen, B., & Suthers, I. M. (2021). The ecology of lepas-based biofouling communities on moored and drifting objects, with applications for marine forensic science. *Marine Biology*, *168*(2), 21. https://doi.org/10.1007/s00227-021-03822-1

Cooper, E. A., Buxey, K. N., Maslen, B. J., & Muhlmann, M. (2020). Retrospective analysis of a bi-national colorectal cancer audit to characterize stage II colon cancer patients who were offered adjuvant chemotherapy. *ANZ Journal of Surgery*, *90*(6), 1136–1140. https://doi.org/10.1111/ans.15735

Folpp, H. R., Schilling, H. T., Clark, G. F., Lowry, M. B., Maslen, B., Gregson, M., & Suthers, I. M. (2020). Artificial reefs increase fish abundance in habitat-limited estuaries. *Journal of Applied Ecology*, *57*(9), 1752–1761. https://doi.org/10.1111/1365-2664.13666

Radford, C., McNutt, J. W., Rogers, T., Maslen, B., & Jordan, N. (2020). Artificial eyespots on cattle reduce predation by large carnivores. *Communications Biology*, *3*(1), 430. https://doi.org/10.1038/s42003-020-01156-0